m-res Numerical Analysis and Statistical Mechanics

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Abstract

Conventional numerical solution of stiff differential equations can require millions of tiny, roundoff-plagued steps. The difficulty disappears if continuous real numbers are discarded in favor of jets of discrete Iris real and m-resolution numbers. Stiff transients vanish in a single step at the resolution threshold, and the remaining response is evaluated in closed algebraic form without division.

Meanwhile, using Iris real and m-res numbers for statistical mechanics gives discrete lattices and maximum entropy as fundamental tools. Matter is matter and not pinpricks, and probabilities are not a phase-space haze but instead represent plausible inferences. On the discrete lattice, maximum entropy over finite configurations yields the Boltzmann distribution directly through contact topology and finite-duration interactions, uniting computation and statistical mechanics.

Applying m-res statistical mechanics to the Master Field Equation predicts viscosity and resistivity from first principles.

Foundations of m-res Numerical Analysis

Conventional numerical analysis is historically abstracted backwards: an unmeasurable, uncountably infinite "continuum" is postulated as if it were known to exist in the external world, and finite discrete computations are denigrated as mere "approximations" plagued by truncation and roundoff errors. However, the only reason to believe a "continuum" exists is conventional mathematics *has words for it*. There is no empirical evidence for such a thing, and the "continuum" provably brings numerous pathologies into mathematics.

In the Iris number system, realities are represented instead at discrete, finite spatial and temporal resolutions. The discrete lattice $\mathcal{G}_N$ and its finite resolution l_{res} provide a language in which it is impossible for the fantasy of a "continuum" to interfere with logical reasoning.

Example 1. Definition: m-Resolution Lattice Mesh and Metric Resolution

Let $\mathcal{G}_N^{(m)} \subset \mathbb{Q}^d$ denote the m-resolution spatial grid in d spatial dimensions defined by a finite resolution increment $\Delta x_{(m)} = 2^{-m} l_0$, where l_0 is the coarse grid spacing and $m \in \mathbb{N}_0$ designates the finite Vernier resolution level. The spatial domain is bounded within a finite box $\Omega = [-L, L]^d$ such that the total number of discrete grid locations is strictly finite:

$$N_{(m)} = \left(\frac{2L}{\Delta x_{(m)}} + 1 \right)^d = \left(\frac{2L}{l_0} 2^m + 1 \right)^d < \infty$$

Each discrete lattice position is addressed by an integer coordinate vector $\mathbf{n} = (n_1, \ldots, n_d) \in \mathbb{Z}^d$ yielding the position vector:

$$\mathbf{x}_{\mathbf{n}} = \mathbf{n} \Delta x_{(m)} \in \mathcal{G}_N^{(m)}$$

Example 2. Definition: Discrete Multivector Difference Operators

Let $\Phi : \mathcal{G}_N^{(m)} \to \mathbb{G}$ be a discrete multivector field configuration in Conformal Geometric Algebra.[1] The discrete spatial directional difference operator $\Delta_k^{(m)}$ along the

spatial basis vector e_k ($k \in \{1, 2, 3\}$) is defined by the central rational difference:

$$\Delta_k^{(m)} \Phi(\mathbf{x}) = \frac{\Phi(\mathbf{x} + \Delta x_{(m)} e_k) - \Phi(\mathbf{x} - \Delta x_{(m)} e_k)}{2\Delta x_{(m)}}$$

The discrete multivector gradient operator on $\mathcal{G}_N^{(m)}$ is the directed sum:

$$D_{(m)} \Phi(\mathbf{x}) = \sum_{k=1}^{d} e_k \Delta_k^{(m)} \Phi(\mathbf{x})$$

which is an exact, rational multivector mapping on $\mathcal{G}_N^{(m)}$ without appeal to limits $\Delta x_{(m)} \to 0$.

Example 3. Theorem: Vernier Stability and Exact Finite Convergence

Let $\Phi^{(m)}$ and $\Phi^{(m+1)}$ denote discrete multivector field configurations on nested grids $\mathcal{G}_N^{(m)} \subset \mathcal{G}_N^{(m+1)}$, and let $\mathcal{P}_m^{m+1} : \mathcal{G}_N^{(m+1)} \to \mathcal{G}_N^{(m)}$ denote the canonical projection operator. If the multivector field satisfies the bounded variation condition:

$$\max_{\mathbf{x} \in \mathcal{G}_N^{(m)}} \left\| \Phi^{(m+1)}(\mathbf{x}) - \Phi^{(m)}(\mathbf{x}) \right\|_{\mathcal{F}} \leq C \cdot 2^{-(m\alpha)}$$

for positive constants $C < \infty$ and $\alpha \geq 1$, then for any operational tolerance $\epsilon_{\text{res}} = 2^{-M}$, there exists a finite integer resolution level:

$$m^* = \left\lceil \frac{1}{\alpha} \log_2 \left(\frac{2C}{\epsilon_{\text{res}}} \right) \right\rceil < \infty$$

such that for all resolution levels $m \geq m^*$, the field values are identically stable within the specified tolerance:

$$\left\| \Phi^{(m)}(\mathbf{x}) - \Phi^{(m^*)}(\mathbf{x}) \right\|_{\mathcal{F}} \leq \epsilon_{\text{res}}$$

without requiring an unphysical infinite-resolution limit.

Proof:

1. Consider the telescopic expansion of the multivector difference across succes-

sive nested resolution levels:

$$\Phi^{(m)}(\mathbf{x}) - \Phi^{(m^*)}(\mathbf{x}) = \sum_{j=m^*}^{m-1} \left(\Phi^{(j+1)}(\mathbf{x}) - \Phi^{(j)}(\mathbf{x}) \right)$$

2. Applying the triangle inequality for the multivector norm in Conformal Geometric Algebra:

$$\left\| \Phi^{(m)}(\mathbf{x}) - \Phi^{(m^*)}(\mathbf{x}) \right\|_{\mathcal{J}} \leq \sum_{j=m^*}^{m-1} \left\| \Phi^{(j+1)}(\mathbf{x}) - \Phi^{(j)}(\mathbf{x}) \right\|_{\mathcal{J}}$$

3. Substituting the bounded variation hypothesis:

$$\sum_{j=m^*}^{m-1} \left\| \Phi^{(j+1)}(\mathbf{x}) - \Phi^{(j)}(\mathbf{x}) \right\|_{\mathcal{J}} \leq C \sum_{j=m^*}^{m-1} 2^{-j\alpha} = C \cdot 2^{-m^*\alpha} \sum_{k=0}^{m-1-m^*} 2^{-k\alpha}$$

4. For $\alpha \geq 1$, the geometric series sum satisfies $\sum_{k=0}^{\infty} 2^{-k\alpha} = (1 - 2^{-\alpha})^{-1} \leq 2$. Therefore:

$$\left\| \Phi^{(m)}(\mathbf{x}) - \Phi^{(m^*)}(\mathbf{x}) \right\|_{\mathcal{J}} \leq 2C \cdot 2^{-m^*\alpha}$$

5. Setting $2C \cdot 2^{-m^*\alpha} \leq \epsilon_{\text{res}}$ and taking logarithms base 2 yields:

$$m^* \geq \frac{1}{\alpha} \log_2 \left(\frac{2C}{\epsilon_{\text{res}}} \right)$$

which is an explicit, finite integer. At resolution level m^*, the computational state is mathematically exact and invariant under further Vernier refinement relative to the resolution ϵ_{res}. This completes the proof.

Example 4. Definition: Genuine Graded m-res Numerical Types and Composite Jets

To formulate a truly novel numerical analysis that departs from classical floating-point approximations, we establish a strict *Gaigen*-style graded algebraic typing discipline.[2] Terminology is kept mathematically sharp: a number of resolution order k consists strictly and exclusively of terms of grade k. A structure containing terms of mixed resolution orders is designated an *m-resolution jet* (or composite jet),

whereas an *m-resolution number* is strictly of a single resolution order.

The foundational types are stratified as follows:

- *Grade 0 (Macroscopic Observable)*: Denoted `real_t`, representing macroscopic scalar observables $x_0 \in \mathbb{R}$ on the grid.
- *Grade 1 (Pure First-Order m-res Number)*: Denoted `mres1_t`, representing pure first-order resolution quantities $x_1\mu$, possessing no macroscopic real component.
- *Grade 2 (Pure Second-Order m-res Number)*: Denoted `mres2_t`, representing pure second-order quantities $x_2\mu^2$.
- *Grade 3 (Pure Third-Order m-res Number)*: Denoted `mres3_t`, representing pure third-order quantities $x_3\mu^3$.
- *Composite Resolution Jet*: Denoted $\texttt{mjet}_K$, aggregating distinct homogeneous grades up to order K:

  $$X = x_0 + x_1\mu + x_2\mu^2 + \cdots + x_K\mu^K$$

  where μ is the canonical resolution generator satisfying discrete lattice nilpotency $\mu^{K+1} \equiv 0$.

Example 5. Theorem: Graded Cauchy Convolution and Exact Nilpotent Inversion

Let $A = \sum_{j=0}^{K} a_j\mu^j$ and $B = \sum_{k=0}^{K} b_k\mu^k$ be composite m-resolution jets in the truncated polynomial ring $\mathbb{R}[\mu]/\langle\mu^{K+1}\rangle$.

1. *Graded Multiplication*: Pure homogeneous grades multiply strictly according to grade addition:

   $$\text{mres}_j * \text{mres}_k \to \text{mres}_{j+k}$$

   For composite jets, multiplication is given by the exact discrete Cauchy convolution:

   $$A * B = \sum_{n=0}^{K} c_n\mu^n, \quad \text{where } c_n = \sum_{j=0}^{n} a_j b_{n-j}$$

All product terms of order $n \geq K+1$ vanish identically by discrete lattice nilpotency.

2. *Exact Nilpotent Inversion*: If $a_0 \neq 0$, the inverse A^{-1} exists and is computed in closed algebraic form without limits, iterative approximations, or subtractive cancellation:

$$A^{-1} = \frac{1}{a_0}(1+\delta)^{-1} = \frac{1}{a_0}\sum_{m=0}^{K}(-1)^m \delta^m$$

where $\delta = \sum_{j=1}^{K} \frac{a_j}{a_0}\mu^j$ is strictly nilpotent ($\delta^{K+1} \equiv 0$).

Proof:

1. The Cauchy convolution follows directly from the distributive law over the graded basis $\{\mu^0, \mu^1, \ldots, \mu^K\}$ modulo the discrete lattice ideal $\langle \mu^{K+1} \rangle$.

2. Because δ has no Grade-0 scalar component, δ^m has minimum grade m. Consequently, for $m \geq K+1$, $\delta^m \equiv 0$. The Neumann series $\sum_{m=0}^{K}(-1)^m\delta^m$ terminates after exactly $K+1$ algebraic terms. Multiplying $(1+\delta)\cdot\sum_{m=0}^{K}(-1)^m\delta^m = 1+(-1)^K\delta^{K+1} = 1$. Multiplying by a_0^{-1} yields the exact multiplicative inverse in closed form. This completes the proof.

Example 6. Definition: Typed Downarrow Projection Operators

In m-res numerical analysis, computation proceeds in the graded jet ring, after which results are projected back to observable states via typed *downarrow operators* $\downarrow$:

1. *Observable Projection* $\downarrow_0$:

$$\downarrow_0 (X) = x_0$$

Extracts the macroscopic Grade-0 observable, discarding sub-resolution Vernier components.

2. *Discrete Grid Projection* $\downarrow_{grid}$:

$$\downarrow_{\text{grid}} (X, \Delta x) = \text{round}_{\mathcal{G}_N}\left(\sum_{k=0}^{K} x_k (\Delta x)^k\right)$$

Evaluates the jet at physical lattice increment Δx and snaps deterministically

to the nearest discrete lattice node of $\mathcal{G}_N$.

3. *Differential Rate Extraction* $\downarrow_k$:

$$\downarrow_k (X) = k! \cdot x_k$$

Extracts the exact k-th order rate of variation without dividing by any small step size Δt, completely avoiding floating-point subtractive cancellation and $0/0$ indeterminate forms.

Digression: The Relationship Between Jets and Taylor's Theorem

To clarify the constructive foundations of m-res numerical analysis for readers unfamiliar with differential algebra or jet geometry, we examine the direct structural relationship between Taylor's theorem and finite graded jets.

Taylor's theorem is fundamentally an exact statement about a finite polynomial of degree K accompanied by an explicit error term (remainder):

$$f(t + \Delta t) = \sum_{k=0}^{K} \frac{f^{(k)}(t)}{k!} (\Delta t)^k + R_K(\Delta t)$$

In authentic numerical analysis, Taylor's theorem is formulated in this operational manner: an exact expression consisting of a finite number of evaluated polynomial terms with the error-bounding remainder as the final term. It does not require an infinite series, nor does it require any continuum.[3] It is an operational decomposition of a local variation into a finite polynomial of degree K and a bounded residual R_K.

However, in classical numerical computation on digital machines, practitioners often discard the polynomial structure and instead approximate individual derivatives using finite difference quotients:

$$\frac{f(t + \Delta t) - f(t)}{\Delta t}$$

When Δt is made small in an effort to minimize truncation error, subtractive cancellation rapidly destroys precision, creating ill-conditioned numerical instability and $0/0$ indeterminate forms.

A *jet* (introduced geometrically by Charles Ehresmann and formalized algebraically

in differential algebra) provides the natural, intrinsic representation of Taylor's theorem. Rather than treating Δt as a numerical increment whose powers must be repeatedly calculated, a jet isolates the finite polynomial itself into a structured algebraic bundle:

$$X = x_0 + x_1\mu + x_2\mu^2 + \cdots + x_K\mu^K$$

where μ is a formal resolution generator whose powers keep the rates of variation separated into distinct, unmixed grades. In this ring, the remainder term R_K is represented algebraically by the discrete nilpotency condition:

$$\mu^{K+1} \equiv 0$$

which reflects the physical reality of the discrete lattice $\mathcal{G}_N$: variations beyond the K-th resolution order are sub-resolution fluctuations that vanish identically into the lattice threshold.

The explicit relationship between the jet coordinates x_k and the derivatives $f^{(k)}$ is determined by the progression of lowering degrees under repeated differentiation:

1. *Descent of Degrees by Differentiation*: Consider an elementary monomial μ^k. Each successive differentiation lowers the algebraic exponent by one and multiplies by the current power:

   $$\frac{d}{d\mu}(\mu^k) = k\mu^{k-1}$$

   $$\frac{d^2}{d\mu^2}(\mu^k) = k(k-1)\mu^{k-2}$$

   Differentiating k times reduces the power to $\mu^0 = 1$, accumulating the descending factorial product:

   $$\frac{d^k}{d\mu^k}(\mu^k) = k \cdot (k-1) \cdot (k-2) \cdots 1 = k!$$

   Any subsequent differentiation of this term yields zero identically.

2. *The Taylor Factorial Denominator*: If a polynomial function $P(\mu) = \sum_{k=0}^{K} x_k\mu^k$ is differentiated k times and evaluated at $\mu = 0$, all terms with degree less than k

have vanished to zero, and all terms with degree greater than k retain at least one factor of μ (which evaluates to zero at $\mu = 0$). The only surviving term is the k-th term:

$$\left.\frac{d^k P}{d\mu^k}\right|_{\mu=0} = k! \cdot x_k$$

In order for the polynomial's k-th rate of change to match the derivative $f^{(k)}(t)$ given by Taylor's theorem, the coordinate x_k must be the normalized Taylor coefficient:

$$x_k = \frac{f^{(k)}(t)}{k!}$$

3. *Exact Polynomial Algebra*: In the graded jet ring $\mathbb{R}[\mu]/\langle\mu^{K+1}\rangle$, operations (addition, Cauchy multiplication, and nilpotent series inversion) proceed purely by polynomial algebra. When two jets multiply, their components combine according to the discrete Cauchy convolution:

$$c_n = \sum_{j=0}^{n} a_j b_{n-j}$$

which automatically computes the exact product-rule derivatives of all orders up to K simultaneously. No step sizes are introduced, and no limits are evaluated.

4. *Recovery of Derivatives via Downarrow Extraction*: When a computation in the jet ring is complete, the physical rates of variation are extracted by the typed downarrow operator $\downarrow_k$:

$$\downarrow_k (X) = k! \cdot x_k = k!\left(\frac{f^{(k)}(t)}{k!}\right) = f^{(k)}(t)$$

The multiplication by $k!$ simply inverts the factorial divisor that arose from the progression of descending exponents during differentiation. This extracts the exact k-th physical rate of variation in closed algebraic form directly from the jet components, without division by small increments and without subtractive cancellation.

Example 7. Theorem: Exact Algebraic Solution of Stiff Systems and Singular Perturbations

Consider a stiff differential equation describing boundary layers, relaxation processes, or reaction networks:

$$\frac{dy}{dt} = -\lambda(y - g(t)), \quad \lambda \gg 1, \quad y(0) = y_0$$

In classical numerical analysis, explicit stepping schemes explode into catastrophic instability unless $\Delta t < 2/\lambda$, forcing millions of tiny steps that accumulate roundoff error.

In m-res numerical analysis, the stiffness is reparameterized by the pure first-order resolution generator $\epsilon = \mu/\lambda$. Then:

1. *Fast Transient Decomposition*: Across a macroscopic step $\Delta t \gg 1/\lambda$, the transient component decays exponentially onto the discrete lattice:

   $$y_{\text{transient}}(t + \Delta t) = (y(t) - g(t)) \exp(-\lambda \Delta t)$$

   When $\lambda \Delta t \geq 50$, the transient falls below the resolution threshold and is rounded to zero on $\mathcal{G}_N$ in a single Vernier step.

2. *Closed-Form Solution*: The driven response is obtained in *closed algebraic form* in the graded m-res jet without time stepping:

   $$Y_{\text{closed}}(t) = \sum_{k=0}^{K} (-1)^k g^{(k)}(t)\, \epsilon^k$$

   The state at macroscopic time $t + \Delta t$ is evaluated unconditionally stably by downarrow projection:

   $$y(t + \Delta t) = y_{\text{transient}}(t + \Delta t) + \downarrow_\epsilon Y_{\text{closed}}(t + \Delta t)$$

   yielding unconditional stability for arbitrarily large macroscopic time increments Δt.

Proof:

1. In the singular perturbation formulation, write the ODE as $\epsilon \frac{dy}{dt} = -(y - g(t))$,

where $\epsilon = \frac{1}{\lambda}\mu$. For the persistent driven response where $\frac{dy}{dt}$ is bounded, expanding $Y = \sum_{k=0}^{K} y_k \epsilon^k$ and equating coefficients of ϵ^k yields the algebraic recurrence:

$$y_0 = g(t), \quad y_k = -y'_{k-1} = (-1)^k g^{(k)}(t)$$

This series is exact up to order K.

2. On the discrete lattice $\mathcal{G}_N$, any transient term with magnitude less than the discrete resolution 2^{-M} is operationally indistinguishable from zero. For $\lambda\Delta t \geq 50$, $\exp(-\lambda\Delta t) \leq \exp(-50) < 2 \cdot 10^{-22}$, which is strictly smaller than double-precision machine epsilon. Therefore, downarrow projection to $\mathcal{G}_N$ evaluates the transient to zero in a single step without numerical oscillation.

3. The closed-form driven component is computed purely through graded jet arithmetic without dividing by Δt. Consequently, the solution remains perfectly bounded, well-conditioned, and physically stable regardless of the magnitude of λ. This completes the proof.

Foundations of m-res Statistical Mechanics

In conventional physics, statistical mechanics commits the following errors:

1. It treats probability distributions as clouds of ethereal substance dispersing itself through some sort of infinite "phase space," despite there being no experimental evidence for such things.

2. It fantasizes matter as point particles with zero volume undergoing an infinite number of collisions. However, if one knows the structure of matter, then this structure must be accounted for. At the very least, matter has spatial extent and cannot undergo infinite numbers of collisions!

In m-resolution statistical mechanics, we adhere strictly to the foundational insights of E. T. Jaynes:[4] probability distributions are not physical fluids; they are explicit, deductive representations of our state of knowledge given incomplete macroscopic information. Phase space is not a continuum of uncountably infinite points, but a finite combinatorial grid of discrete state cells partitioned by the fundamental spatial and momentum resolution scales $(\Delta x_{(m)}, \Delta p_{(m)})$. Furthermore, phase space is a linguistic entity, not a physical one. It would not exist if no one ever thought it up and wrote it down.

The order of abstracting is correct: phase space is recognized as *language that talks about* physical realities, whereas physical realities *are not and cannot be words*.[5]

Example 8. Definition: Discrete m-Resolution Phase Space Cell Network

Let an isolated physical system composed of K localized field packets reside within a finite box Ω on the discrete lattice $\mathcal{G}_N^{(m)}$. Let the momentum of each field packet be bounded within a finite multivector momentum range $\mathcal{P} = [-P_{\max}, P_{\max}]^d$ with discrete momentum resolution $\Delta p_{(m)} = 2^{-m} p_0$. The one-particle phase space is partitioned into a finite number of discrete cells:

$$W_1 = \left(\frac{2L}{\Delta x_{(m)}}\right)^d \times \left(\frac{2P_{\max}}{\Delta p_{(m)}}\right)^d < \infty$$

For K indistinguishable discrete field configurations, the total number of microstates

Ω_{micro} is strictly finite and combinatorial:

$$\Omega_{\text{micro}} = \binom{W_1 + K - 1}{K} < \infty$$

Every microstate $i \in \{1, 2, \dots, \Omega_{\text{micro}}\}$ corresponds to an exact, non-singular configuration of discrete field cells on $\mathcal{G}_N^{(m)}$.

Example 9. Definition: Discrete Maximum Entropy (MaxEnt) State Inference

Let p_i denote the probability assigned to discrete microstate $i \in \{1, \dots, \Omega_{\text{micro}}\}$, satisfying normalization:

$$\sum_{i=1}^{\Omega_{\text{micro}}} p_i = 1, \quad p_i \geq 0$$

Given a set of M macroscopic expectation constraints $\langle A_r \rangle = \sum_{i=1}^{\Omega_{\text{micro}}} p_i A_r(i) = \bar{A}_r$ for $r \in \{1, \dots, M\}$, the Jaynesian information entropy over the discrete cell network is:

$$S_I = -\sum_{i=1}^{\Omega_{\text{micro}}} p_i \ln p_i$$

The uniquely honest deductive probability assignment that incorporates all macroscopic constraints without introducing unwarranted assumptions is the distribution maximizing S_I:

$$p_i = \frac{1}{Z} \exp\left(-\sum_{r=1}^{M} \lambda_r A_r(i)\right)$$

where the partition function Z is the finite discrete sum:

$$Z(\lambda_1, \dots, \lambda_M) = \sum_{i=1}^{\Omega_{\text{micro}}} \exp\left(-\sum_{r=1}^{M} \lambda_r A_r(i)\right)$$

and λ_r are Lagrange multipliers determined deductively by:

$$-\frac{\partial \ln Z}{\partial \lambda_r} = \bar{A}_r$$

Statistical Mechanical Reformulation of the Master Field Equation

In the Iris unified field theory, the fundamental law of physics is the Master Field Equation:

$$DF = J$$

formulated in Conformal Geometric Algebra (CGA) over canonical basis $\{e_1, e_2, e_3, e_4, e_\odot, e_\infty\}$ with real-valued components. Here, F is the complete multivector field containing scalar, vector, bivector, trivector, and pseudoscalar components, D is the geometric derivative, and J is the multivector source current. This Master Field Equation amounts to Maxwell's equations with geometric constraints required by the Michelson-Morley experimental result. From it can be derived not General Relativity, which is incorrect, but instead Gauss's theory of gravity. The two "nuclear forces" are found to be mistakes.

We wish to treat this Master Field Equation as a statistical mechanical system.

In statistical mechanics done correctly, one cannot simply postulate ad hoc Brownian motion, random Langevin noise forces, or phenomenological friction terms. Instead, one must *make deductions* about the motions of particles and field kernels from first principles:

1. Localized mass-energy concentrations are discrete field packets (vortices and solitons of F).
2. The mechanical acceleration, trajectory, and momentum transfer of any localized entity are deduced directly from the surface integral of the multivector Poynting-stress energy-momentum tensor over the boundary enclosing the entity.
3. Every influence propagates at finite speed c, so all interactions involve causally delayed boundary fluxes across the discrete grid.
4. When microscopic detail is unknown, the macroscopic current J and macroscopic stress tensor are deduced via Maximum Entropy expectation over the discrete lattice states.

Example 10. Definition: Multivector Energy-Momentum Flux Tensor

Let F be the multivector electromagnetic-gravitational field satisfying $DF = J$ in Conformal Geometric Algebra on the discrete lattice $\mathcal{G}_N^{(m)}$. The symmetric multivector energy-momentum-stress tensor $T(a)$ acting on an arbitrary directional vector a is defined by:

$$T(a) = -\frac{1}{2}\left\langle FaF\right\rangle_1$$

where $\widetilde{F}$ denotes the geometric reversion of F, and $\langle ... \rangle_1$ denotes the vector projection in Conformal Geometric Algebra. The local divergence of T on the discrete lattice satisfies the exact conservation law:

$$D \cdot T = \langle J\widetilde{F}\rangle_1$$

representing the direct contact momentum and energy transfer between field fluxes and localized currents.

Example 11. Conclusion Pending New Information: Deductive Derivation of Particle Trajectory from Boundary Flux Balance

Let a localized hypothetical object or particle occupy a discrete spatial domain $\Omega_k \subset \mathcal{G}_N^{(m)}$ bounded by the discrete surface boundary $\partial\Omega_k$. Let $P_k = \sum_{\mathbf{x}\in\Omega_k} \rho_{\text{field}}(\mathbf{x})\Delta x_{(m)}^d$ denote the net multivector field momentum contained within Ω_k. Then the time rate of change of momentum $\Delta P_k / \Delta t$ of the particle is not a postulated Newton's second law, but is rigorously deduced from the surface flux of the multivector stress tensor:

$$\frac{\Delta P_k}{\Delta t} = -\sum_{\mathbf{s}\in\partial\Omega_k} T(n(\mathbf{s}))\Delta A(\mathbf{s}) + \sum_{\mathbf{x}\in\Omega_k} \langle J_{\text{internal}}\widetilde{F}\rangle_1 \Delta x_{(m)}^d$$

where $n(\mathbf{s})$ is the outward directed unit normal vector on $\partial\Omega_k$ and $\Delta A(\mathbf{s})$ is the discrete boundary cell area.

Furthermore, because electromagnetic influences travel across the lattice at finite speed c, waves emitted by distant sources require finite time to reach the boundary $\partial\Omega_k$. Thus, the field $F(\mathbf{x}, t)$ acting on the surface at time t depends on what surround-

ing sources were doing at the earlier moment:

$$t' = t - \frac{\|\mathbf{x} - \mathbf{x}'\|}{c}$$

when their influence was first sent out.

Deduction:

1. On the discrete lattice $\mathcal{G}_N^{(m)}$, sum the discrete divergence of the multivector stress tensor T over all cells inside Ω_k:

$$\sum_{\mathbf{x}\in\Omega_k} (D \cdot T)(\mathbf{x}) \Delta x_{(m)}^d = \sum_{\mathbf{x}\in\Omega_k} \langle J\widetilde{F}\rangle_1(\mathbf{x}) \Delta x_{(m)}^d$$

2. Applying the discrete Gauss-Ostrogradsky divergence identity on $\mathcal{G}_N^{(m)}$, the volume sum of the divergence transforms identically into the discrete surface flux sum over the boundary $\partial\Omega_k$:

$$\sum_{\mathbf{x}\in\Omega_k} (D \cdot T)(\mathbf{x}) \Delta x_{(m)}^d = \sum_{\mathbf{s}\in\partial\Omega_k} T(n(\mathbf{s})) \Delta A(\mathbf{s})$$

3. The total current J decomposes into localized internal binding current J_{internal} and external convective/field current. The time rate of change of the enclosed momentum within the finite cell domain Ω_k is:

$$\frac{\Delta P_k}{\Delta t} = \sum_{\mathbf{x}\in\Omega_k} \frac{\Delta}{\Delta t} \left(\frac{1}{c^2} \mathbf{S}(\mathbf{x}) \right) \Delta x_{(m)}^d$$

   where $\mathbf{S} = \langle F e_4 \widetilde{F} \rangle_1$ is the Poynting momentum flux density.

4. Equating the internal time rate of momentum change with the net balance of incoming/outgoing boundary stress and internal current forces:

$$\frac{\Delta P_k}{\Delta t} = -\sum_{\mathbf{s}\in\partial\Omega_k} T(n(\mathbf{s})) \Delta A(\mathbf{s}) + \sum_{\mathbf{x}\in\Omega_k} \langle J_{\text{internal}} \widetilde{F} \rangle_1 \Delta x_{(m)}^d$$

5. Because all wave packet interactions travel at finite speed c, the field values $F(\mathbf{s})$ arriving at the boundary $\partial\Omega_k$ from any other source k' at distance $r_{kk'}$ require

finite propagation time:

$$\Delta t_{\text{prop}} = \frac{r_{kk'}}{c} > 0$$

The acceleration and subsequent displacement $\Delta \mathbf{x}_k = \frac{p_k}{M_k} \Delta t$ of the object are thus uniquely deduced from the field stress balance without introducing unphysical zero-duration action-at-a-distance or arbitrary kinematic axioms. This completes the deduction.

Example 12. Conclusion Pending New Information: MaxEnt Reformulation of the Master Field Equation over Discrete Ensembles

Let an ensemble of microscopic field kernels and current trajectories be distributed over $\mathcal{G}_N^{(m)}$ according to the Jaynesian MaxEnt probability distribution p_i. Let $\langle F(\mathbf{x}) \rangle = \sum_i p_i F_i(\mathbf{x})$ and $\langle J(\mathbf{x}) \rangle = \sum_i p_i J_i(\mathbf{x})$ denote the macroscopic MaxEnt expectation fields. Then the macroscopic Master Field Equation satisfies:

$$D_{(m)} \langle F(\mathbf{x}) \rangle = \langle J(\mathbf{x}) \rangle$$

and the macroscopic energy-momentum transport equation deduced from the ensemble average contains an exact, non-zero viscous dissipation tensor:

$$D_{(m)} \cdot \langle T \rangle = \langle J \rangle \cdot \langle F \rangle + \mathbf{C}_{\text{visc}}$$

where the correlation tensor $\mathbf{C}_{\text{visc}}$ arises entirely from microscopic covariance:

$$\mathbf{C}_{\text{visc}}(\mathbf{x}) = \sum_{i=1}^{\Omega_{\text{micro}}} p_i \left(J_i(\mathbf{x}) - \langle J(\mathbf{x}) \rangle \right) \left(\widetilde{F}_i(\mathbf{x}) - \langle \widetilde{F}(\mathbf{x}) \rangle \right)$$

which is identically the physical origin of viscosity, electrical resistivity, and thermal relaxation in macroscopic matter, deduced without phenomenological damping constants.

Deduction:

1. The microscopic Master Field Equation holds in every discrete microstate configuration $i \in \{1, \dots, \Omega_{\text{micro}}\}$ on $\mathcal{G}_N^{(m)}$:

$$D_{(m)} F_i(\mathbf{x}) = J_i(\mathbf{x})$$

2. Taking the linear expectation value with respect to the discrete MaxEnt probability distribution p_i:

$$\sum_{i=1}^{\Omega_{\text{micro}}} p_i D_{(m)} F_i(\mathbf{x}) = \sum_{i=1}^{\Omega_{\text{micro}}} p_i J_i(\mathbf{x})$$

3. Because the discrete difference operator $D_{(m)}$ is linear and independent of the ensemble weights p_i, it commutes with the expectation sum:

$$D_{(m)} \left(\sum_{i=1}^{\Omega_{\text{micro}}} p_i F_i(\mathbf{x}) \right) = \sum_{i=1}^{\Omega_{\text{micro}}} p_i J_i(\mathbf{x}) \implies D_{(m)} \langle F(\mathbf{x}) \rangle = \langle J(\mathbf{x}) \rangle$$

proving the macroscopic linearity of the field equation.

4. Now consider the quadratic energy-momentum stress divergence equation in each microstate:

$$D_{(m)} \cdot T_i(\mathbf{x}) = \langle J_i(\mathbf{x}) \widetilde{F}_i(\mathbf{x}) \rangle_1$$

5. Taking the MaxEnt expectation value of both sides:

$$D_{(m)} \cdot \langle T(\mathbf{x}) \rangle = \sum_{i=1}^{\Omega_{\text{micro}}} p_i \langle J_i(\mathbf{x}) \widetilde{F}_i(\mathbf{x}) \rangle_1$$

6. Decomposing the microscopic variables into mean and fluctuating components:

$$J_i(\mathbf{x}) = \langle J(\mathbf{x}) \rangle + \delta J_i(\mathbf{x})$$

$$F_i(\mathbf{x}) = \langle F(\mathbf{x}) \rangle + \delta F_i(\mathbf{x})$$

where $\sum_i p_i \delta J_i = 0$ and $\sum_i p_i \delta F_i = 0$.

7. Expanding the product inside the expectation sum:

$$\sum_{i=1}^{\Omega_{\text{micro}}} p_i \left(\langle J \rangle + \delta J_i \right) \left(\langle \widetilde{F} \rangle + \delta \widetilde{F}_i \right) = \langle J \rangle \langle \widetilde{F} \rangle + \sum_{i=1}^{\Omega_{\text{micro}}} p_i \delta J_i \delta \widetilde{F}_i$$

8. Defining $\mathbf{C}_{\text{visc}}(\mathbf{x}) \equiv \sum_{i=1}^{\Omega_{\text{micro}}} p_i \delta J_i(\mathbf{x}) \delta \widetilde{F}_i(\mathbf{x})$, we obtain:

$$D_{(m)} \cdot \langle T(\mathbf{x}) \rangle = \langle J(\mathbf{x}) \rangle \langle \widetilde{F}(\mathbf{x}) \rangle + \mathbf{C}_{\text{visc}}(\mathbf{x})$$

9. The covariance term $\mathbf{C}_{\text{visc}}$ represents the net momentum and energy scattered into unresolved microscopic modes. Because wave packet propagation requires finite duration at speed c, delayed correlations between fluctuating currents and radiated fields are strictly dissipative:

$$\sum_{\mathbf{x} \in \mathcal{G}_N^{(m)}} \mathbf{C}_{\text{visc}}(\mathbf{x}) \cdot \langle \mathbf{v} \rangle \geq 0$$

This is the rigorous first-principles derivation of macroscopic shear viscosity, bulk viscosity, and Ohm's law resistivity. Every macroscopic dissipative parameter is deduced from the underlying field dynamics and incomplete state information, without ever postulating phenomenological friction. This completes the deduction.

Conclusion

We have established a unified, constructive formulation of numerical analysis and statistical mechanics rooted in the Iris number system:

1. *m-res Numerical Analysis*: Computational algorithms on discrete lattices $\mathcal{G}_N$ are recognized as fundamental and exact operators, replacing illusory continuum limits with finite Vernier stability bounds. By replacing continuous real numbers with graded m-resolution jets, stiff transients vanish at the resolution threshold in a single step, and the remaining response is evaluated in closed algebraic form without division.

2. *m-res Statistical Mechanics*: Phase space is partitioned into finite combinatorial cells, and probability distributions are established strictly as Jaynesian deductive inferences derived from Maximum Entropy subject to operational constraints.

3. *Deductive Field Reformulation*: When applied to the Master Field Equation $DF = J$, the motions of particles are rigorously deduced from the surface flux of the multivector stress tensor and causally delayed wave propagation, while macroscopic dissipative laws (viscosity, resistivity) emerge deductively as the covariance of unresolved microscopic fluctuations. By eliminating unphysical continuum infinities and fictitious action-at-a-distance, physical and numerical mechanics are united in a single, operational deductive discipline.

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1. Leo Dorst, Daniel Fontijne, and Stephen Mann. 2007. *Geometric Algebra for Computer Science: An Object-Oriented Approach to Geometry*. Morgan Kaufmann Publishers, San Francisco, CA. ISBN: 978-0-12-369461-4. In this paper we extend Conformal Geometric Algebra with a sixth, null-magnitude basis vector representing the time dimension.

2. Daniel Fontijne. 2006. Gaigen 2: a geometric algebra implementation generator. In Proceedings of the 5th international conference on Generative programming and component engineering (GPCE '06). Association for Computing Machinery, New York, NY, USA, 141–150. https://doi.org/10.1145/1173706.1173728

3. Brook Taylor published this relation in his 1715 treatise *Methodus Incrementorum*

Directa et Inversa, investigating the calculus of finite differences and finite increments. The widespread modern academic belief that Taylor's theorem is an "infinite sum" requiring uncountable continuum limits is an anachronism: Taylor died in 1731, more than a century before the set-theoretic continuum dogmas of Dedekind and Cantor were conceived. In numerical analysis, Taylor's theorem establishes the exact existence of a remainder number for any chosen finite number of terms, serving directly as an operational error bound.

4. Edwin T. Jaynes. 2003. *Probability Theory: The Logic of Science.* Cambridge University Press, Cambridge, UK. [Edited by G. Larry Bretthorst].

5. Alfred Korzybski. 1941. *Science and Sanity: An Introduction to Non-Aristotelian Systems and General Semantics (2nd ed.).* International Non-Aristotelian Library Publishing Company. Be aware much of the science related in *Science and Sanity* is incorrect. For instance, Korzybski should have recognized immediately that Special and General Relativity involve reversals of the correct order of abstracting, thus must be wrong. *A change of coordinate system cannot affect the physical world!* The Unified Field Theory that Einstein never found is simply Maxwell's equations, but accounting for mass-energy equivalence (about which Einstein was correct). Rather than recite word magic, we must *explain* Michelson-Morley. *The reason no "ether wind" was detected is Earth and the Michelson-Morley apparatus themselves are* part of *the "ether"*—what we more properly call *the universal electromagnetic field.*

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